THE ERDŐS IMPERATIVE: BERTRAND'S POSTULATE AS A ROUTE TO THE TWIN PRIME CONJECTURE

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Channel Exhaustion (CE) implies the Twin Prime Conjecture (TPC) via Bertrand's Postulate, without appeal to the Prime Number Theorem. The proof has been formally verified in Lean 4 with zero `sorry`, using Bertrand's Postulate as a Mathlib theorem. The single remaining ingredient for a full proof of TPC is a proof of CE itself.[1]

Contents

Part 1. What Has Been Proved

1. Introduction

The Erdős Imperative is a methodological principle: replace every appeal to the Prime Number Theorem (PNT) in [5] with the most elementary statement that suffices. Paul Erdős, who gave the first elementary proof of the prime number theorem [3], would have insisted on it.

The present paper applies this imperative to the route from the Bridging Conjecture (BC) to the Twin Prime Conjecture (TPC). The route suggested in [5] invokes Euclid's theorem (primes are infinite) as the final contradiction. We show that the same channel exhaustion mechanism delivers a sharper, more constructive statement: an explicit interval $[n, 2n]$ containing no prime, directly contradicting Bertrand's Postulate.

Date: May 2026.

Bertrand's Postulate admits proofs requiring only elementary combinatorics [2]. The route is therefore fully elementary, from first principles to TPC, conditional on one conjecture: Channel Exhaustion.

The Lean 4 formalisation in Section 5 makes this precise. The axiom audit at the end of that section confirms: three standard axioms, zero `sorry`, Bertrand from Mathlib.

2. Notation and Background

We follow the notation of [5]. A *twin prime witness* is an integer $w \geq 1$ such that $6w-1$ and $6w+1$ are both prime. We write $\mathcal{W}$ for the set of all twin prime witnesses (OEIS A002822).

Lemma 2.1 (Primes in channels, [5, Lemma 2.3])**.** *Every prime $p > 3$ satisfies $p = 6m-1$ or $p = 6m+1$ for some $m \geq 1$.*

Conjecture 2.2 (Channel Exhaustion, CE)**.** *If $\mathcal{W}$ is finite with largest element $w_{\max}$, then for every $m > w_{\max}$, both $6m-1$ and $6m+1$ are composite.*

Theorem 2.3 (Bertrand's Postulate [1, 2])**.** *For every integer $n \geq 1$, there exists a prime p with $n < p \leq 2n$.*

3. The Main Theorem

Theorem 3.1 (CE $\Rightarrow$ $\mathcal{W}$ finite $\Rightarrow \neg$BP)**.** *Assume CE. If $\mathcal{W}$ is finite with largest element $w_{\max}$, then Bertrand's Postulate fails: the interval $(6w_{\max}+1,\ 12w_{\max}+2]$ contains no prime.*

Proof. Let $n = 6w_{\max}+1$. Let p be any prime with $p > n$. Since $p > 3$, Lemma 2.1 gives $m \geq 1$ with $p = 6m-1$ or $p = 6m+1$. Since $p > 6w_{\max}+1$, we have $m > w_{\max}$. By CE, both $6m-1$ and $6m+1$ are composite — contradicting p prime. Hence no prime exceeds n, and the Bertrand interval $(n, 2n] = (6w_{\max}+1, 12w_{\max}+2]$ is prime-free. □

Remark 3.2 (The prime-free interval is a lower bound). Theorem 3.1 establishes that the prime-free region extends *at least* to $12w_{\max}+2$. This is a structural lower bound derived from the channel representation alone. The true extent of the prime-free region is expected to be larger.

The Denaturing Machine (DM) — the variant of the Bridging Machine that discards twin prime pairs and sweeps exceptions to the set X — requires time to clear the witness exceptions above $w_{\max}$ before CE fully takes hold. This sweep accounts for additional composite territory beyond the structural minimum. Based on the dynamics of the DM, the extinction zone is conjectured to lie in the range

$$(6w_{\max}+1,\ cw_{\max}], \qquad 12 \leq c \leq 24,$$

with the true value of c an open question to be settled computationally.

Corollary 3.3 (CE $\Rightarrow$ TPC)**.** *Assuming CE, the Twin Prime Conjecture holds.*

Proof. By contrapositive of Theorem 3.1: CE and $\mathcal{W}$ finite jointly imply $\neg$BP. Since BP is a theorem, CE and $\mathcal{W}$ finite cannot both hold. Therefore CE implies $\mathcal{W}$ is infinite, which is TPC. □

Remark 3.4. The step $\neg$BC $\Rightarrow \mathcal{W}$ finite (via the Bridging Machine halting) is definitional and does not appear in the Lean proof: the formal proof works directly with $\mathcal{W}$ finite as a hypothesis and derives $\neg$BP, making BC unnecessary as an intermediate step. The chain reduces to: CE $\Rightarrow$ TPC.

5. Lean 4 Formalisation

The following code was produced by Aristotle (Anthropic's specialised Lean 4 coding agent) in session `20260530T072226Z`. All four theorems carry zero `sorry`. Bertrand's Postulate is used as `Nat.exists_prime_lt_and_le_two_mul` from Mathlib.

```
/-- The lower channel value: L_m = 6m - 1 -/
def lowerChannel (m : N) : N := 6 * m - 1

/-- The upper channel value: H_m = 6m + 1 -/
def upperChannel (m : N) : N := 6 * m + 1

/-- A positive integer m is a witness if both 6m-1 and 6m+1 are prime. -/
def IsWitness (m : N) : Prop :=
  1 <= m /\ (6 * m - 1 : N).Prime /\ (6 * m + 1 : N).Prime

/-- Channel Exhaustion: if all witnesses lie in S, then beyond S
    both channel values are composite. -/
def ChannelExhaustion : Prop :=
  forall (S : Finset N), (forall m, IsWitness m -> m in S) ->
    forall m, (forall s in S, s < m) ->
      not (6 * m - 1 : N).Prime /\ not (6 * m + 1 : N).Prime

/-- The Twin Prime Conjecture. -/
def TPC : Prop := forall N : N, exists m, m > N /\ IsWitness m



-- Key lemma: CE makes all primes bounded by 6 * max(S) + 1
theorem no_prime_above_bound
    (hce : ChannelExhaustion)
    (S : Finset N) (hS : forall m, IsWitness m -> m in S) :
    forall p, Nat.Prime p -> p <= 6 * S.sup id + 1

-- Main theorem: CE + W finite => Bertrand violated
-- (establishes lower bound; true extinction zone conjectured 12-24 * w_max)
theorem ce_wfinite_implies_not_bertrand
    (hce : ChannelExhaustion)
    (hfin : (setOf IsWitness).Finite) :
    exists n : N, n /= 0 /\
      not exists p, Nat.Prime p /\ n < p /\ p <= 2 * n

-- Corollary: CE => W infinite (using Mathlib Bertrand)
theorem bp_ce_implies_w_infinite
```

```
    (hce : ChannelExhaustion) :
    not (setOf IsWitness).Finite

-- Full chain: CE => TPC
theorem bp_ce_implies_tpc
    (hce : ChannelExhaustion) : TPC
```

5.3. **Axiom audit.**

```
#print axioms bp_ce_implies_tpc
-- propext, Classical.choice, Quot.sound
```

Three standard axioms. No `sorry`. No custom axioms.

Acknowledgements

The author thanks Claude (Anthropic) for collaboration throughout this programme, for the phrase "channel exhaustion," and for the observation that the Erdős Imperative applies here. Aristotle (Anthropic) produced the Lean 4 formalisation in Part 1. The question "why grandparents and not parents?" was posed in a 3 am conversation and has not yet been answered.

See [6] for full AI attribution.

References

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[4] F. Balestrieri, Some observations on the twin prime conjecture, *arXiv*:1106.6050, 2011.

[5] J. Seymour, *A Structural Hypothesis for the Infinitude of Twin Primes*, Independent Researcher, Wild Duck Theories, Sydney, Australia, 2026. https://wildducktheories.github.io/twin-primes/papers/structural/wdt-structural-hypothesis.pdf

[6] J. Seymour, *How AI Was Used: A Transparency Statement*, Independent Researcher, Wild Duck Theories, Sydney, Australia, 2026. https://wildducktheories.github.io/twin-primes/papers/ai-usage/wdt-ai-usage.pdf

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